

CHAPTER 9

GENERAL RELATIVITY: THE FIELD THEORY APPROACH

We move now to the modern approach to General Relativity: field theory. The chief advantage of this formulation is that it is simple and easy; the only thing to specify is the so-called Lagrangian density. We start by presenting a simple introduction to classical field theory in flat spacetime which we later generalize to curved spacetime. The last part of the Chapter is devoted to the action for the gravitational field and the recovery of Einstein equations from it.

9.1 Classical mechanics

The fundamental problem of classical mechanics is to determine the way particles move given a potential. Dynamical systems can be described by equations of motion or by action functionals

$$S = \int dt L(q_j(t), \dot{q}_j(t)) \quad (9.1)$$

depending on generalized coordinates and velocities $\{q_j(t), \dot{q}_j(t)\}$. The classical trajectory is defined as the unique path that extremizes the action functional ($\delta S = 0$) for all variations $q_j \rightarrow q_j + \delta q_j$ with fixed initial $q_j(t_0)$ and final values $q_j(t_f)$. An explicit variation of the action gives

$$\delta S = \int_{t_0}^{t_f} dt \left(\frac{\partial L}{\partial q_j} \delta q_j + \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \right). \quad (9.2)$$

Integrating the last term by parts to flip the temporal derivative onto $\partial L / \partial \dot{q}_j$ we get

$$\delta S = \int_{t_0}^{t_f} dt \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \right] \delta q_j, \quad (9.3)$$

where we have omitted a total derivative that vanishes because of the boundary conditions $\delta q_j(t_0) = \delta q_j(t_f) = 0$. Since δq_j is arbitrary, the extremization of the action translates into the so-called Euler-Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_j} = 0. \quad (9.4)$$

This variational formulation has several advantages:

- i) The properties of the system are compactly summarized in one function, the Lagrangian.
- ii) There is a direct connection between invariances of the Lagrangian and constants of motion¹.
- iii) There is a close relation between the Lagrangian formulation of classical mechanics and quantum mechanics.

9.2 From Classical Mechanics to Field theory

The Lagrangian formalism presented above can be extended to continuous systems involving an infinite number of degrees of freedom. This is achieved by taking the appropriate limit of a system with a finite number of degrees of freedom. Consider a one dimensional chain of length l made of N equal masses m . The masses are separated by a distance a and connected by identical massless springs with force constant k . The total length of the system is $l = (N + 1)a$. The displacement of the particles with respect to its equilibrium position $\bar{x}_j = ja$ is described by a generalized coordinate $\phi(x_j, t) \equiv x_j(t) - \bar{x}_j$ with $j = 1, \dots, N$ and $\phi_0 = \phi_{N+1} = 0$. The Lagrangian of the full system includes the kinetic energy of the particles and the energy stored into the springs, i.e.

$$L = T - U = \frac{1}{2}m \sum_{j=1}^N \dot{\phi}_j^2(t) - \frac{1}{2}k \sum_{j=0}^N (\phi_{j+1}(t) - \phi_j(t))^2. \quad (9.5)$$

The continuous limit of the previous expression can be taken by sending $N \rightarrow \infty$ and $a \rightarrow 0$ in such a way that the total length of the chain, $l = (N + 1)a$, remains fixed. To keep the total mass of the system and the force between particles finite we require m/a and ka to go to some finite values μ and Y playing the role of the mass density and the Young modulus in the continuous theory. We have

$$L = \frac{1}{2} \sum_{j=1}^N a \left(\frac{m}{a} \right) \dot{\phi}_j^2(t) - \frac{1}{2} \sum_{j=0}^N a (ka) \left(\frac{\phi_{j+1}(t) - \phi_j(t)}{a} \right)^2 \longrightarrow L = \frac{1}{2} \int_0^l dx \left[\mu \dot{\phi}^2 - Y (\partial_x \phi)^2 \right],$$

with the finite number of generalized coordinates ϕ_j replaced by a continuous function $\phi(x, t)$. The antisymmetric dependence of Eq.(9.6) on the derivatives suggests the introduction of a set of coordinates $x^\mu = (c_s t, x)^T$ with $c_s = \sqrt{Y/\mu}$ and a Lorentzian metric $\eta_{\mu\nu} = \text{diag}(-1, 1)$. This allows us to write

$$S = \int d(c_s t) dx \mathcal{L} \quad (9.6)$$

with

$$\mathcal{L} = -\frac{\mu c_s}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi, \quad (9.7)$$

the so-called *Lagrangian density*. The jump from fields existing within a physical medium to fields in vacuum is now straightforward: we must simply replace c_s by the speed of light c . Generalizing the metric $\eta_{\mu\nu}$ to arbitrary dimensions, we can write generically the action for relativistic fields as

$$S = \int d^n x \mathcal{L}(\phi, \partial_\mu \phi), \quad (9.8)$$

where we have allowed for a dependence of the Lagrangian density on the fields.



Exercise

Consider again the chain of masses connected by springs. Modify the system to give rise to an explicit dependence of the Lagrangian on ϕ . Hint: Eq. (9.7) is shift-invariant.

¹For instance, if the Lagrangian is invariant under rotations, angular momentum is conserved.



Gauge freedom in the Lagrangian

The Lagrangian of a physical system is not unique. To see this, consider a transformation of the form

$$\mathcal{L} \longrightarrow \mathcal{L}' = \mathcal{L} + \partial_\sigma g^\sigma, \quad (9.9)$$

and its effect on the action (9.8)

$$S' = \int d^n x \mathcal{L}' = S + \int_{\mathcal{R}} d^n x \partial_\sigma g^\sigma = S + \int_{\partial \mathcal{R}} g^\sigma dS_\sigma. \quad (9.10)$$

The term associated to $\partial_\sigma g^\sigma$ turns out to be a boundary term, which does not contribute to the equations of motion. Lagrangians differing by a contribution $\partial_\sigma g^\sigma$ give rise to the same equations of motion.

The equations of motion for the field $\phi(x, t)$ can be obtained by considering the change of the action under an infinitesimal change $\phi(x, t) \longrightarrow \phi(x, t) + \delta\phi(x, t)$. The only requirement to be satisfied by the variations $\delta\phi$ is to be differentiable and to vanish outside some bounded region of spacetime (to allow an integration by parts). Performing this variation we get

$$\delta S = \int d^n x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial \phi_{,\mu}} \delta\phi_{,\mu} \right] \delta\phi = \int d^n x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \right] \delta\phi. \quad (9.11)$$

Requiring the action to be stationary ($\delta S = 0$) and taking into account that $\delta\phi$ is completely arbitrary, we obtain the continuous version of the Euler-Lagrange equations

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0. \quad (9.12)$$



A worked-out example

As a direct application of Eq. (9.12), let me consider the action (9.7)

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0 \quad \longrightarrow \quad \partial_\mu (\eta^{\mu\nu} \partial_\nu \phi) = 0 \quad \longrightarrow \quad -\frac{1}{c_s^2} \frac{\partial^2 \phi}{\partial t^2} + \frac{\partial^2 \phi}{\partial x^2} = 0. \quad (9.13)$$

As expected, we get a wave equation.

9.3 Principles of Lagrangian construction

What kind of Lagrangian density should we choose? To be in the safe side, the Lagrangian density of a *relativistic theory* is required to satisfy the following requirements:

1. $\mathcal{L}$ must be a real-valued function, since it enters into expressions of physical significance, like the Hamiltonian.
2. $\mathcal{L}$ must have dimension 4 in units of energy, since in natural units the action is dimensionless and $[d^4 x] = -4$.
3. $\mathcal{L}$ must be a linear combination of Lorentz invariant quantities constructed from the fields, their first partial derivatives and the universally available objects $\eta_{\mu\nu}$ and $\epsilon_{\mu\nu\rho\sigma}$.

4. The coefficients of this linear combination can be restricted by the symmetries of the problem (internal symmetries/ gauge symmetries).
5. $\mathcal{L}$ should be bounded from below.

The power of the previous program is made most vividly evident by considering some examples.

9.3.1 A complex scalar field with $U(1)$ symmetry

Consider a complex scalar field

$$\Phi = \phi_1 + i\phi_2, \quad \Phi^* = \phi_1 - i\phi_2. \quad (9.14)$$

The quadratic Lorentz invariants which can be constructed from $\Phi, \Phi^*, \partial_\mu \Phi$ and $\partial_\mu \Phi^*$ lead to a Lagrangian density of the form

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} [a \partial_\mu \Phi \partial_\nu \Phi + a^* \partial_\mu \Phi^* \partial_\nu \Phi^* + 2a_0 \partial_\mu \Phi^* \partial_\nu \Phi] + \frac{1}{2} [b \Phi \Phi + b^* \Phi \Phi^* + 2b_0 \Phi^* \Phi]. \quad (9.15)$$

where the reality condition $\mathcal{L} = \mathcal{L}^*$ imposes the appearance of the pairs a, a^* and b, b^* and requires the coefficients a_0 and b_0 to be real. The previous Lagrangian density can be written in a more compact way by introducing the arrays

$$\tilde{\Phi} \equiv \begin{pmatrix} \Phi \\ \Phi^* \end{pmatrix} \quad \text{and} \quad \tilde{\Phi}^\dagger \equiv \begin{pmatrix} \Phi^* \\ \Phi \end{pmatrix}^T = (\Phi^* \ \Phi). \quad (9.16)$$

to get²

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} \tilde{\Phi}^\dagger_{,\mu} \begin{pmatrix} a_0 & a^* \\ a & a_0 \end{pmatrix} \tilde{\Phi}_{,\nu} + \frac{1}{2} \tilde{\Phi}^* \begin{pmatrix} b_0 & b^* \\ b & b_0 \end{pmatrix} \tilde{\Phi}. \quad (9.17)$$

The number of terms appearing in this Lagrangian can be reduced in cases in which we have symmetries on top of Lorentz invariance. As an illustration of this, imagine the field Φ to possess an internal symmetry

$$\Phi \rightarrow e^{i\omega} \Phi, \quad \Phi^* \rightarrow e^{-i\omega} \Phi^*. \quad (9.18)$$

In this case, we necessarily have $a = b = 0$ and the matrices in (9.17) become diagonal. This leaves us with a simpler Lagrangian, that with some notational adjustments, can be written as

$$\mathcal{L} = \frac{1}{2} K [-\eta^{\mu\nu} \partial_\mu \Phi^* \partial_\nu \Phi - \kappa^2 \Phi^* \Phi]. \quad (9.19)$$

A direct application of the Euler-Lagrange equations (9.12) provides two uncoupled equations for Φ and Φ^* , namely

$$(\square - \kappa^2) \Phi = 0, \quad (\square - \kappa^2) \Phi^* = 0, \quad (9.20)$$

that we can use to provide a physical interpretation for the parameter κ^2 . Indeed, setting $\Phi = e^{ik_\mu x^\mu}$ with $p^\mu = (E, \mathbf{p})$ in any of these two equations, we get a dispersion relation $E^2 - \mathbf{p}^2 = \kappa^2$, which makes it natural to identify the parameter κ^2 with the *mass* m^2 of the field.

²In this notation, the reality condition $\mathcal{L} = \mathcal{L}^*$ results from the hermiticity of the 2×2 matrices.

9.3.2 A worked-out example: Vector and tensor fields

The previous analysis can be easily extended to other kinds of fields. Consider for instance a vector field A_μ and a tensor field $B_{\mu\nu}$. Which is the most general Lagrangian $\mathcal{L}(A, \partial A, B, \partial B)$ that can be constructed out of these two fields? To clarify the procedure, let me split the problem into several pieces

$$\mathcal{L}(A, B, \partial A, \partial B) = \mathcal{L}_A(A, \partial A) + \mathcal{L}_B(B, \partial B) + \mathcal{L}_{\text{int}}(A, B, \partial A, \partial B). \quad (9.21)$$

The lagrangian $\mathcal{L}_A$ is a *free lagrangian* for A_μ , i.e. a linear combination of Lorentz invariant terms quadratic in A_μ ³,

$$\mathcal{L}_A = a_1 \partial_\mu A^\mu \partial_\nu A^\nu + a_2 \partial_\mu A_\nu \partial^\mu A^\nu + a_3 \partial_\mu A_\nu \partial^\nu A^\mu + a_4 A_\mu A^\mu + a_5 \epsilon_{\mu\nu\rho\sigma} (\partial^\nu A^\mu) (\partial^\sigma A^\rho). \quad (9.22)$$

The lagrangian $\mathcal{L}_B$ is a free lagrangian for $B_{\mu\nu}$

$$\begin{aligned} \mathcal{L}_B = & b_1 \partial_\mu B^\nu{}_\nu \partial^\mu B^\kappa{}_\kappa + b_2 \partial_\mu (B^{\kappa\nu} + B^{\nu\kappa}) \partial^\mu (B_{\kappa\nu} + B_{\nu\kappa}) + b_3 \partial_\mu (B^{\mu\nu} + B^{\nu\mu}) \partial_\nu B^\kappa{}_\kappa + \\ & + b_4 \partial_\mu (B^{\kappa\mu} + B^{\mu\kappa}) \partial^\nu (B_{\kappa\nu} + B_{\nu\kappa}) + b_5 \epsilon_{\mu\nu\rho\sigma} B^{\mu\nu} B^{\rho\sigma} + b_6 B_{\mu\nu} B^{\mu\nu} + b_7 B_{\mu\nu} B^{\nu\mu} + b_8 (B^\mu{}_\mu)^2. \end{aligned} \quad (9.23)$$

The interaction Lagrangian $\mathcal{L}_{\text{int}}$ follows the same logic, but now involving the interactions between the two fields,

$$\begin{aligned} \mathcal{L}_{\text{int}} = & i_1 \partial_\mu A_\nu (B^{\mu\nu} + B^{\nu\mu}) + i_2 A_\mu \partial_\nu (B^{\mu\nu} + B^{\nu\mu}) + i_3 \epsilon_{\mu\nu\rho\sigma} (\partial^\mu A^\nu) B^{\rho\sigma} + \\ & + i_4 \epsilon_{\mu\nu\rho\sigma} A^\mu (\partial^\nu B^{\rho\sigma}) + i_5 A_\mu A_\nu B^{\mu\nu}. \end{aligned} \quad (9.24)$$

I maybe missed some terms, but I think you get the idea. As in the scalar case, the list of operators and independent coefficients can be shortened cases in which there are extra symmetries on top on Lorentz invariance. Let me illustrate this with a simpler example.

The action for the electromagnetic field

Consider the action (9.22) alone. The first thing that can simplify our life is the gauge freedom in the choice of the Lagrangian density. In particular notice that choosing $g_\sigma = \epsilon_{\mu\nu\rho\sigma} (\partial^\nu A^\mu) A^\rho$ in (9.9) allows us to eliminate the term a_5 in (9.22), since

$$\partial^\sigma g_\sigma = \epsilon_{\mu\nu\rho\sigma} (\partial^\nu A^\mu) (\partial^\sigma A^\rho) + \underbrace{\epsilon_{\mu\nu\rho\sigma} (\partial^\nu \partial^\sigma A^\mu) A^\rho}_{0 \text{ by symmetry}}. \quad (9.25)$$

Taking this into account we are left with an action containing 4 pieces

$$S = \int d^4x [a_1 \partial_\mu A^\mu \partial_\nu A^\nu + a_2 \partial_\mu A_\nu \partial^\mu A^\nu + a_3 \partial_\mu A_\nu \partial^\nu A^\mu + a_4 A_\mu A^\mu]. \quad (9.26)$$

Imagine now that A_μ is the field of a *gauge theory*. In that case the field configurations A_μ and

$$A'_\mu = A_\mu + \partial_\mu \chi, \quad (9.27)$$

with arbitrary scalar function χ give rise to the same physical observables⁴. This automatically forbids the a_4 term in (9.26)⁵ and puts some restrictions on the other coefficients. To see this, let me split $\partial_\mu A_\nu$ into its symmetric $S_{\mu\nu} \equiv \partial_{(\mu} A_{\nu)}$ and antisymmetric $F_{\mu\nu} \equiv \partial_{[\mu} A_{\nu]}$ parts

$$\partial_\mu A_\nu = S_{\mu\nu} + F_{\mu\nu}, \quad (9.28)$$

³Quadratic actions give rise to linear equations of motion, where the superposition principle can be applied.

⁴In the same way that *physicality* cannot be attributed to $\mathcal{L}$, we cannot make any claim about the physicality of A_μ . Physicality might be attributed to the set $\{A_\mu\}$ of gauge-equivalent 4-potentials or to any gauge invariant attribute of that set, but not to its individual elements.

⁵It cannot be compensated by the transformation of the other (derivative) terms.

and rewrite the action (9.26) as

$$S = \int d^4x [a_1 S_\mu^\mu S^\nu_\nu + (a_2 + a_3) S_{\mu\nu} S^{\mu\nu} + (a_2 - a_3) F_{\mu\nu} F^{\mu\nu}] . \quad (9.29)$$

The invariance of the action under the gauge transformation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$ requires $a_1 = 0$ and $a_3 = -a_2$. This restriction leaves us with an action

$$S = \int d^4x F_{\mu\nu} F^{\mu\nu} , \quad (9.30)$$

where we have omitted an overall normalization factor that can be determined by choosing the coupling of the gauge field A_μ to matter and the units of that coupling. The equations of motion associated with this action can be computed via the Euler-Lagrange equations (9.12) or by varying the action with respect to A_μ . We follow the second procedure to get

$$\begin{aligned} \delta S &= \int d^4x [F^{\mu\nu} \delta F_{\mu\nu} + F_{\mu\nu} \delta F^{\mu\nu}] = 2 \int d^4x F^{\mu\nu} \delta F_{\mu\nu} \\ &= 2 \int d^4x F^{\mu\nu} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) = 4 \int d^4x F^{\mu\nu} \partial_\mu \delta A_\nu \\ &= -4 \int d^4x \partial_\mu F^{\mu\nu} \delta A_\nu + \text{boundary terms} , \end{aligned} \quad (9.31)$$

where we used the symmetry properties of $F_{\mu\nu}$ and performed an integration by parts. Imposing finally the condition $\delta S = 0$ for arbitrary δA_ν , we arrive to the very familiar result

$$\partial_\mu F^{\mu\nu} = 0 . \quad (9.32)$$

The Maxwell equations in vacuum are recovered from an action (9.30) constructed with very limited principles, namely, quadraticity in the fields, Lorentz invariance and gauge invariance.

9.4 The action for the graviton

The procedure outlined in the previous section is quite powerful. As an interesting application for General Relativity, let me consider the action for a second rank symmetric and massless tensor field $h_{\mu\nu}$. As in the vector field case, the kinetic term is constructed out of scalars that are quadratic in the derivatives $\partial_\rho h_{\mu\nu}$. The most general expression will be the sum of the different scalars obtained by contracting pairs of indices in all possible ways. The resulting action takes the form

$$S = \int d^4x [c_1 \partial_\mu h^\nu_\nu \partial^\mu h^\kappa_\kappa + c_2 \partial_\mu h^{\kappa\nu} \partial^\mu h_{\kappa\nu} + c_3 \partial_\mu h^{\mu\nu} \partial_\nu h^\kappa_\kappa + c_4 \partial_\mu h^{\kappa\mu} \partial^\nu h_{\kappa\nu}] , \quad (9.33)$$

with c_1, c_2, c_3, c_4 some undetermined constants. These constants can be determined up to an overall factor by requiring the action to be invariant under the *gauge transformations*

$$h_{\mu\nu} \rightarrow h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu . \quad (9.34)$$

Plugging (9.34) into (9.33) and performing some simple manipulations we get

$$\begin{aligned} S \rightarrow S + \int d^4x [&-2(2c_1 + c_3) \partial_\mu h^\kappa_\kappa \partial^\mu \partial_\lambda \xi^\lambda - 2(2c_2 + c_4) \partial^\mu h_{\mu\nu} \square \xi^\nu - 2(c_3 + c_4) \partial^\mu h_{\mu\nu} \partial^\nu \partial_\kappa \xi^\kappa \\ &+ (4c_1 + 2c_2 + 4c_3 + 3c_4) \partial_\mu \partial_\kappa \xi^\kappa \partial^\mu \partial_\lambda \xi^\lambda + (2c_2 + c_4) \square \xi_\mu \square \xi^\mu] , \end{aligned} \quad (9.35)$$

which imposing $\delta S = 0$, provides the constraints

$$c_2 = -c_1, \quad c_4 = -c_3 = 2c_1. \quad (9.36)$$

Taking this into account, the action (9.33) takes the form

$$S = \int d^4x [\partial_\mu h^{\kappa\nu} \partial^\mu h_{\kappa\nu} + 2\partial_\mu h^{\mu\nu} \partial_\nu h^\kappa_\kappa - 2\partial_\mu h^{\kappa\mu} \partial^\nu h_{\kappa\nu} - \partial_\mu h^\nu_\nu \partial^\mu h^\kappa_\kappa], \quad (9.37)$$

where we have omitted an overall normalization factor that can be determined by specifying the coupling to matter and setting the units of the coupling. The associated equations of motion can be obtained by varying the action with respect to the field. This leads to

$$\begin{aligned} \delta S = \int d^4x [2\partial^\mu h_{\kappa\nu} \partial_\mu \delta h^{\kappa\nu} + 2\partial_\nu h^\kappa_\kappa \partial_\mu \delta h^{\mu\nu} + 2\eta_{\kappa\lambda} \partial_\mu h^{\mu\nu} \partial_\nu \delta h^{\kappa\lambda} \\ - 2\partial^\nu h_{\kappa\nu} \partial_\mu \delta h^{\kappa\mu} - 2\partial^\mu h_{\kappa\mu} \partial_\nu \delta h^{\kappa\nu} - 2\eta_{\kappa\lambda} \partial^\mu h^\nu_\nu \partial_\mu \delta h^{\kappa\lambda}] , \end{aligned} \quad (9.38)$$

which, integrating by parts, dropping boundary terms and renaming indices can be written as

$$\begin{aligned} \delta S = 2 \int d^4x [-\square h_{\kappa\nu} \delta h^{\kappa\nu} - \partial_\mu \partial_\nu h^\kappa_\kappa \delta h^{\mu\nu} - \eta_{\kappa\lambda} \partial_\mu \partial_\nu h^{\mu\nu} \delta h^{\kappa\lambda} + \partial_\mu \partial^\nu h_{\kappa\nu} \delta h^{\kappa\mu} + \partial^\mu \partial_\nu h_{\kappa\mu} \delta h^{\kappa\nu} + \eta_{\kappa\lambda} \square h^\nu_\nu \delta h^{\kappa\lambda}] \\ = 2 \int d^4x [-\square h_{\mu\nu} - \partial_\mu \partial_\nu h^\kappa_\kappa - \eta_{\mu\nu} \partial_\lambda \partial^\lambda h^\kappa_\kappa + \partial_\mu \partial^\kappa h_{\kappa\nu} + \partial_\nu \partial^\kappa h_{\kappa\mu} + \eta_{\mu\nu} \square h^\kappa_\kappa] \delta h^{\mu\nu}. \end{aligned} \quad (9.39)$$

A simple inspection reveals that the quantity inside the square brackets is nothing else than the linearized version of the Einstein tensor $G_{\mu\nu}$

$$\delta S \propto \int d^4x G_{\mu\nu} \delta h^{\mu\nu} . \quad \longrightarrow \quad G_{\mu\nu} = 0 . \quad (9.40)$$

The linearized version of Einstein equations in vacuum are recovered from an action (9.39) constructed with very limited principles, namely, quadraticity in the fields, Lorentz invariance and gauge invariance.

9.5 Field theory in curved spacetime

The variational approach for fields presented in the previous section can be generalized to include the interaction with gravity. We guess the form of the action in this case with the help of the Equivalence Principle:

- Replace the Minkowski metric $\eta_{\mu\nu}$ by $g_{\mu\nu}$.
- Replace partial derivatives by covariant derivatives (*colon-goes-to semicolon rule*).
- Replace the Lorentz invariant volume element $d^n x$ by the covariant volume element $d^n x \sqrt{|g|}$.

Since $\tilde{\mathcal{L}}$ and $d^n x \sqrt{|g|}$ are scalars under general coordinate transformations, the resulting action

$$S = \int d^n x \underbrace{\sqrt{|g|} \tilde{\mathcal{L}}(\phi, \nabla_\mu \phi, g_{\mu\nu})}_{\mathcal{L}} \quad (9.41)$$

is guaranteed to provide covariant equations of motion. Note that the untilded quantity $\mathcal{L} \equiv \sqrt{|g|} \tilde{\mathcal{L}}$ is a scalar density of weight 1.

9.6 The energy-momentum tensor

Consider now an arbitrary *infinitesimal* coordinate transformation

$$x^\mu \rightarrow \bar{x}^\mu = x^\mu + \xi^\mu(x). \quad (9.42)$$

This transformation generates a perturbation to both the fields and the metric in such a way that the Lagrangian density $\mathcal{L}$ (no tilde) becomes

$$\mathcal{L}(\phi + \delta\phi, \phi_{,\mu} + \delta\phi_{,\mu}, g_{\mu\nu} + \delta g_{\mu\nu}) \approx \mathcal{L}(\phi, \phi_{,\mu}, g_{\mu\nu}) + \frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial \phi_{,\mu}} \delta\phi_{,\mu} + \frac{\partial \mathcal{L}}{\partial g_{\mu\nu}} \delta g_{\mu\nu}. \quad (9.43)$$

Integrating by parts, we get two pieces

$$\delta S = \int d^n x \underbrace{\left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \right]}_{=0} \delta\phi + \int d^n x \frac{\partial \mathcal{L}}{\partial g_{\mu\nu}} \delta g_{\mu\nu}. \quad (9.44)$$

The first one is associated to a *particular variation* $\delta\phi$ and vanishes when taking into account the Euler-Lagrange equation for ϕ . The second term must be then equal to zero for S to remain unchanged. The integrand $\partial \mathcal{L} / \partial g_{\mu\nu}$ is a scalar density. Let's define a symmetric second-rank tensor out of such a density

$$T^{\mu\nu} \equiv \frac{2}{\sqrt{|g|}} \frac{\partial \mathcal{L}}{\partial g_{\mu\nu}}, \quad (9.45)$$

and write

$$\delta S = \frac{1}{2} \int d^n x \sqrt{|g|} T^{\mu\nu} \delta g_{\mu\nu}. \quad (9.46)$$

Although it is tempting to simply set $\sqrt{|g|} T_{\mu\nu} = 0$, this condition is overly restrictive, since $\delta g_{\mu\nu}$ refers here to a *specific* type of variation, not to an arbitrary one. The variation $\delta g_{\mu\nu}$ can be however expressed in terms of the arbitrary perturbation ξ_μ by taking into account that $\delta g_{\mu\nu} = -(\xi_{\mu;\nu} + \xi_{\nu;\mu})$ (cf. Eq. (6.60) and notice (9.50)). This gives

$$\begin{aligned} \delta S &= \frac{1}{2} \int d^n x \sqrt{|g|} T^{\mu\nu} \delta g_{\mu\nu} = - \int d^n x \sqrt{|g|} T^{\mu\nu} \xi_{\mu;\nu} \\ &= \int d^n x \sqrt{|g|} T^{\mu\nu}{}_{;\nu} \xi_\mu - \underbrace{\int d^n x \left(\sqrt{|g|} T^{\mu\nu} \xi_\mu \right)_{;\nu}}_{=0}, \end{aligned} \quad (9.47)$$

where we have made use of the symmetry property of $T^{\mu\nu}$ and integrated by parts to get a total derivative that vanishes by assumption on the boundary of integration. Since ξ_μ is arbitrary we must have

$$\nabla_\nu T^{\mu\nu} = 0, \quad (9.48)$$

which is a continuity equation suggesting that we can identify the tensor (9.45) with the energy-momentum tensor of any physical system.

**A common sign mistake**

You will find some books giving an alternative definition of the energy momentum tensor

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\partial \mathcal{L}}{\partial g^{\mu\nu}}, \quad (9.49)$$

in terms of $\delta g^{\mu\nu}$ rather than $\delta g_{\mu\nu}$. The difference in sign between these two equivalent expressions comes from

$$g_{\mu\nu} g^{\nu\lambda} = \delta^\mu_\lambda \quad \rightarrow \quad \delta(g_{\mu\nu} g^{\nu\lambda}) = 0 \quad \rightarrow \quad \delta g^{\mu\nu} = -g^{\mu\lambda} g^{\nu\rho} \delta g_{\lambda\rho}. \quad (9.50)$$

9.6.1 A particular case

When the Lagrangian $\tilde{\mathcal{L}}$ on $\mathcal{L} = \sqrt{|g|}\tilde{\mathcal{L}}$ depends only on the metric and not on the first derivatives of the metric⁶, i.e. $\tilde{\mathcal{L}} = \tilde{\mathcal{L}}(\phi, \partial_\mu \phi, g_{\mu\nu})$, it is possible to derive an alternative expression for the energy-momentum tensor. In particular, taking into account that $\frac{\partial \sqrt{|g|}}{\partial g_{\mu\nu}} = \frac{1}{2} \sqrt{|g|} g^{\mu\nu}$, we can write

$$T^{\mu\nu} = \frac{2}{\sqrt{|g|}} \frac{\partial \mathcal{L}}{\partial g_{\mu\nu}} = \frac{2}{\sqrt{|g|}} \left(\frac{\partial \sqrt{|g|}}{\partial g_{\mu\nu}} \tilde{\mathcal{L}} + \sqrt{|g|} \frac{\partial \tilde{\mathcal{L}}}{\partial g_{\mu\nu}} \right) = \frac{2}{\sqrt{|g|}} \left(\frac{1}{2} \sqrt{|g|} g^{\mu\nu} \tilde{\mathcal{L}} + \sqrt{|g|} \frac{\partial \tilde{\mathcal{L}}}{\partial g_{\mu\nu}} \right), \quad (9.51)$$

or equivalently

$$T^{\mu\nu} = g^{\mu\nu} \tilde{\mathcal{L}} + 2 \frac{\partial \tilde{\mathcal{L}}}{\partial g_{\mu\nu}}. \quad (9.52)$$

**Exercise**

Compute the energy-momentum tensor for (9.7).

9.7 The Einstein-Hilbert action

In order to construct an action for the gravitational field we must define a Lagrangian. This Lagrangian must transform as a scalar under general coordinate transformations and depend on the metric tensor and its derivatives. Note however that gravity is completely different from all other fundamental interactions, since non-trivial quantity can be constructed from the metric and its *first derivatives* alone. This can be easily seen by considering an arbitrary scalar combination $f(g_{\mu\nu}, \partial_\rho g_{\mu\nu})$ of this quantities around a small region in spacetime. According to the local flatness theorem, in such a region, it is always possible to find coordinates such that $g_{\mu\nu} = \eta_{\mu\nu}$ and $\partial_\rho g_{\mu\nu} = 0$ and therefore $f = \text{constant}$. But, since we are dealing with a scalar quantity, this will be the case in any other coordinate system. In other words, any covariant scalar function constructed just from the metric tensor and its derivatives will be a trivial constant.

The simplest non-trivial quantity that can be constructed from the metric and its derivatives is the Ricci scalar, which depends on the metric and its first and second order derivatives. The resulting action reads

$$S_{EH} = \int d^4x \mathcal{L}_{EH} = \frac{1}{2\kappa^2} \int d^4x \sqrt{|g|} R = \frac{1}{2\kappa^2} \int d^4x \sqrt{|g|} g^{\mu\nu} R_{\mu\nu} \quad (9.53)$$

⁶Particular cases are the lagrangian for scalar fields or that for the electromagnetic field, where the covariant derivative reduces to the standard derivative.

and is known as the *Einstein-Hilbert action*. The constant of proportionality κ^2 is included on dimensional grounds and will be determined at the end of the computation.

**Exercise**

Which is the dimension of κ^2 ?

Consider the variation of (9.53) resulting from the variation of the metric tensor

$$g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}, \quad (9.54)$$

where $\delta g_{\mu\nu}$ and its first derivative are assumed to vanish at the boundary of the integration region. We obtain

$$\delta \mathcal{L}_{EH} \propto \underbrace{\sqrt{|g|} \delta g^{\mu\nu} R_{\mu\nu} + \delta \sqrt{|g|} R}_{\delta \mathcal{L}_1} + \underbrace{\sqrt{|g|} g^{\mu\nu} \delta R_{\mu\nu}}_{\delta \mathcal{L}_2}, \quad (9.55)$$

where we have defined two pieces, $\mathcal{L}_1$ and $\mathcal{L}_2$. The first one can be easily evaluated by taking into account the variation and the variation of the metric determinant

$$\delta \sqrt{|g|} = -\frac{1}{2} \sqrt{|g|} g_{\mu\nu} \delta g^{\mu\nu}. \quad (9.56)$$

**Exercise**

Prove Eq. (9.56).

Collecting the terms, we get

$$\delta \mathcal{L}_1 = \sqrt{|g|} \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \delta g^{\mu\nu}. \quad (9.57)$$

The evaluation of $\delta \mathcal{L}_2$ is slightly more involved since it requires to perform the variation of the Ricci tensor. The easiest way of doing this is to consider the variation of the Riemann tensor and perform the required contractions at the end of the computation. Schematically, the variation of the Riemann tensor has the structure

$$\text{Riemann} \sim \partial \Gamma + \Gamma \Gamma \quad \longrightarrow \quad \delta(\text{Riemann}) \sim \delta(\partial \Gamma) + (\delta \Gamma) \Gamma, \quad (9.58)$$

so the first thing that we have to compute is the variation of Christoffel symbols $\delta \Gamma_{\mu\nu}^\rho$ defined by

$$\Gamma_{\mu\nu}^\rho \xrightarrow{g_{\mu\nu} + \delta g_{\mu\nu}} \tilde{\Gamma}_{\mu\nu}^\rho = \Gamma_{\mu\nu}^\rho + \delta \Gamma_{\mu\nu}^\rho \quad \Longrightarrow \quad \delta \Gamma_{\mu\nu}^\rho = \tilde{\Gamma}_{\mu\nu}^\rho - \Gamma_{\mu\nu}^\rho. \quad (9.59)$$



Be careful! We are just performing a variation of the metric, not transforming it.

It is easy to see that, even though a connection is *not* a tensor, the difference of two connections $\delta \Gamma_{\mu\nu}^\rho$ transforms as a tensor, i.e.

$$\delta \bar{\Gamma}_{\mu\nu}^\rho = \frac{\partial \bar{x}^\rho}{\partial x^\sigma} \frac{\partial x^\lambda}{\partial \bar{x}^\mu} \frac{\partial x^\kappa}{\partial \bar{x}^\nu} \delta \Gamma_{\lambda\kappa}^\rho. \quad (9.60)$$

**Exercise**

Check it.

The property (9.60) extremely simplifies the computation of the variation of the Riemann tensor. Indeed, we can always go to a local free fall reference frame in which $\Gamma_{\mu\nu}^\rho = 0$ at some arbitrary point P . In such a point the expression (9.58) becomes

$$\delta R_{\mu\nu} = \partial_\rho(\delta\Gamma_{\mu\nu}^\rho) - \partial_\nu(\delta\Gamma_{\rho\mu}^\rho) = \nabla_\rho(\delta\Gamma_{\mu\nu}^\rho) - \nabla_\nu(\delta\Gamma_{\rho\mu}^\rho), \quad (9.61)$$

where in the last step used of the fact that the partial and covariant derivatives coincide when $\Gamma_{\mu\nu}^\rho = 0$. The resulting *Palatini equation*

$$\delta R_{\mu\nu} = \nabla_\rho(\delta\Gamma_{\mu\nu}^\rho) - \nabla_\nu(\delta\Gamma_{\rho\mu}^\rho) \quad (9.62)$$

is a *tensorial equation* (remember the property (9.60)) valid in *any arbitrary coordinate system* (and *not only* in the free fall reference frame at P).

**Exercise**

Prove that the second term in the previous expression is symmetric, as it should be.

A similar trick can be applied to get the explicit expression of $\delta\Gamma$, that takes the same form that the definition of the Christoffel symbols, with the metric replaced by the metric variation and the partial derivatives replaces by covariant derivatives, i.e.

$$\delta\Gamma_{\nu\rho}^\mu = \frac{1}{2}g^{\mu\sigma}(\nabla_\nu\delta g_{\sigma\rho} + \nabla_\rho\delta g_{\sigma\nu} - \nabla_\sigma\delta g_{\nu\rho}). \quad (9.63)$$

**Exercise**

Check the previous expression by explicit computation.

Taking (9.62) into account, $\delta\mathcal{L}_2$ becomes

$$\delta\mathcal{L}_2 = \sqrt{|g|} g^{\mu\nu} [\nabla_\rho(\delta\Gamma_{\mu\nu}^\rho) - \nabla_\nu(\delta\Gamma_{\rho\mu}^\rho)] = \sqrt{|g|} \nabla_\sigma [g^{\mu\nu}\delta\Gamma_{\mu\nu}^\sigma - g^{\mu\sigma}\delta\Gamma_{\rho\mu}^\rho], \quad (9.64)$$

where we have used the metric compatibility condition (4.68). We are left therefore with the covariant divergence of a vector. Using the property

$$V^\mu{}_{;\mu} = \frac{1}{\sqrt{|g|}} \left(\sqrt{|g|} V^\mu \right)_{;\mu}, \quad (9.65)$$

we get the boundary term

$$\delta\mathcal{L}_2 = \sqrt{|g|} \nabla_\sigma [g^{\mu\nu}\delta\Gamma_{\mu\nu}^\sigma - g^{\mu\sigma}\delta\Gamma_{\rho\mu}^\rho] = \partial_\sigma \left[\sqrt{|g|} g^{\mu\nu} (\delta\Gamma_{\mu\nu}^\sigma) - \sqrt{|g|} g^{\mu\sigma} (\delta\Gamma_{\rho\mu}^\rho) \right]. \quad (9.66)$$



As I said before, gravity is a quite particular field theory. The existence of *second derivatives* in the Einstein-Hilbert action gives rise to a contribution depending on the value of the first derivatives on the boundary. To deal with these, we have two options:

- Extend the variational principle and require the fields *and* their derivatives to be fixed at the boundary. This would give rise to reasonable field equations. A clear example from classical mechanics illustrating this would be

$$S = \int_{t_0}^{t_f} dt \left(\ddot{q} + \frac{1}{2} \dot{q}^2 \right) = \dot{q}(t_f) - \dot{q}(t_0) + \frac{1}{2} \int_{t_0}^{t_f} \dot{q}^2, \quad (9.67)$$

with the assumption that *both* $\dot{q}$ and q are fixed at the boundary. This approach has however some caveats. On the one hand, it does not obey a composition rule of the kind

$$S(0 \rightarrow 1 \rightarrow 2) = S(0 \rightarrow 1) + S(1 \rightarrow 2), \quad (9.68)$$

where the paths connecting (q_0, t_0) and (q_2, t_2) are decomposed at an intermediate time t_1 with $t_0 < t_1 < t_2$. Although the paths are expected to be continuous at $t = t_1$, they do not need to be smooth at that point which requires leaving $\dot{q}_1$ free at $t = t_1$. On the other hand, the action principle has its roots in quantum mechanics, where the simultaneous fixing of q and $\dot{q}$ is inappropriate.

- Add the so-called Gibbons-Hawking-York counterterm to the action

$$S = S_{EH} + S_{HGY} = \frac{1}{2\kappa^2} \int_{\mathcal{R}} d^4x \sqrt{|g|} R + \frac{1}{\kappa^2} \int_{\partial\mathcal{R}} d^3x \sqrt{|h|} K, \quad (9.69)$$

with h the determinant of the induced metric on the boundary and K the trace of the extrinsic curvature. The Gibbons-Hawking-York is constructed in such a way that its variation cancels the unwanted term associated to the second derivatives of the metric, keeping only the part associated to the quadratic part of the action. Proving this statement is beyond the scope of this course. The interested reader is referred to the excellent discussion in Padmanabhan's book.

Forgetting about the boundary term, the Einstein-Hilbert action (9.53) becomes

$$\delta S_{EH} = \frac{1}{2\kappa^2} \int d^4x \sqrt{|g|} \left[R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right] \delta g^{\mu\nu}, \quad (9.70)$$

which, demanding it to vanish for arbitrary variations $\delta g^{\mu\nu}$, gives us the Einstein's equations in the absence of matter

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0. \quad (9.71)$$

9.8 Einstein equations in the presence of matter

Having obtained the Einstein equations in the vacuum, let us now derive its full form in the presence of matter. Consider the action

$$S = S_{EH} + S_M, \quad (9.72)$$

with S_M containing all the non-gravitational fields. The variation of S_M with respect to $\delta g^{\mu\nu}$ (upper indices) gives

$$\delta S_M = -\frac{1}{2} \int d^4x \sqrt{|g|} T_{\mu\nu} \delta g^{\mu\nu}, \quad (9.73)$$

where we have made use of the covariant definition (9.49). Putting everything together we get

$$\delta S_{EH} + \delta S_M = \frac{1}{2\kappa^2} \int d^4x \sqrt{|g|} \left[R_{\mu\nu} - \frac{1}{2} R - \kappa^2 T_{\mu\nu} \right] \delta g^{\mu\nu}. \quad (9.74)$$

Since $\delta g^{\mu\nu}$ is arbitrary, we must have

$$G_{\mu\nu} = \kappa^2 T_{\mu\nu}, \quad (9.75)$$

which confirms the identification of (9.45), or (9.49), with the energy-momentum tensor and allows us to identify the proportionality constant κ^2 with $8\pi G$.



Exercise

Modify the Einstein-Hilbert action to obtain the Einstein equations with cosmological constant.